Detailed Backend Roadmap for AI-Powered IDE Project

1. Python

Why We Use It: Python is the core backend language used for AI integration, APIs, authentication, automation, and server-side logic.

Topics to Learn Deeply:
• Functions and OOP
• Async programming
• File handling
• Error handling
• Modules and packages
• Virtual environments
• Decorators

Best Source: https://docs.python.org/3/tutorial/

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2. FastAPI

Why We Use It: FastAPI is used to create fast backend APIs for AI requests, authentication, project management, and terminal operations.

Topics to Learn Deeply:
• Routing
• Request and response models
• Pydantic
• Dependency injection
• Async APIs
• Middleware
• API documentation

Best Source: https://fastapi.tiangolo.com/

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3. REST APIs

Why We Use It: REST APIs connect frontend and backend for communication and data transfer.

Topics to Learn Deeply:
• GET, POST, PUT, DELETE
• Status codes
• JSON handling
• Authentication
• Error responses

Best Source: https://restfulapi.net/

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4. Authentication & Authorization

Why We Use It: Authentication secures the IDE by managing login systems, protected APIs, and user permissions.

Topics to Learn Deeply:
• JWT tokens
• OAuth basics
• Password hashing
• Role-based access
• Session handling

Best Source: https://jwt.io/introduction

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5. MongoDB

Why We Use It: MongoDB stores user data, project files, AI chats, settings, and IDE configurations.

Topics to Learn Deeply:
• Collections and documents
• CRUD operations
• Indexes
• Aggregation
• Relationships
• MongoDB Atlas

Best Source: https://www.mongodb.com/docs/

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6. SQL Databases

Why We Use It: SQL databases are useful for structured data like billing, analytics, and user management.

Topics to Learn Deeply:
• Tables
• Joins
• Normalization
• Queries
• Indexes
• Relationships

Best Source: https://www.w3schools.com/sql/

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7. WebSockets

Why We Use It: WebSockets enable real-time communication for AI streaming, collaborative coding, and live terminal updates.

Topics to Learn Deeply:
• Socket connections
• Live events
• Streaming responses
• Bi-directional communication

Best Source: https://websockets.readthedocs.io/

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8. Docker

Why We Use It: Docker isolates environments and helps run user code securely inside containers.

Topics to Learn Deeply:
• Docker images
• Containers
• Dockerfile
• Volumes
• Container networking

Best Source: https://docs.docker.com/get-started/

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9. Linux & Terminal

Why We Use It: Linux knowledge is important for backend deployment, server management, and terminal execution.

Topics to Learn Deeply:
• Basic commands
• Permissions
• Processes
• Shell scripting
• Package management

Best Source: https://linuxjourney.com/

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10. Redis

Why We Use It: Redis is used for caching, session storage, rate limiting, and real-time processing.

Topics to Learn Deeply:
• Caching
• Pub/Sub
• Session management
• Data expiration

Best Source: https://redis.io/docs/

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11. AI API Integration

Why We Use It: AI API integration connects your backend with AI models like ChatGPT, code generation, and debugging systems.

Topics to Learn Deeply:
• OpenAI APIs
• Streaming responses
• Prompt handling
• Context management
• Rate limits

Best Source: https://platform.openai.com/docs

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12. File Management

Why We Use It: File management handles user project files, uploads, downloads, and code storage.

Topics to Learn Deeply:
• Read/write operations
• Directory handling
• Temporary storage
• File uploads

Best Source: https://docs.python.org/3/library/os.html

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13. Background Tasks & Queues

Why We Use It: Background tasks process long-running jobs like AI generation and project builds.

Topics to Learn Deeply:
• Celery basics
• Task queues
• Async workers
• Scheduling

Best Source: https://docs.celeryq.dev/en/stable/

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14. Security

Why We Use It: Security protects APIs, user data, terminal access, and backend infrastructure.

Topics to Learn Deeply:
• CORS
• Rate limiting
• Input validation
• SQL injection prevention
• Environment variables

Best Source: https://owasp.org/www-project-top-ten/

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15. Deployment & DevOps

Why We Use It: Deployment makes the backend accessible online and ensures scalability.

Topics to Learn Deeply:
• Render deployment
• AWS basics
• CI/CD
• Nginx
• Environment setup

Best Source: https://docs.github.com/en/actions